PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Association between splenectomy and chronic thromboembolic pulmonary hypertension: A systematic review and meta-analysis
AUTHORS	Zhang, Liyan; Pei Jing, Yan; Yang, Kehu; Wu, Shanlian; Bai, Yuping; Zhu, Xin-Yu; Chen, Xiaojie; Li, Li; Cao, Yunshan; Zhang, Min

VERSION 1 – REVIEW

REVIEWER	Marco C. Post Departments of Cardiology St. Antonius Hospital, Nieuwegein/Utrecht University Medical Center Utrecht The Netherlands
REVIEW RETURNED	04-May-2020

GENERAL COMMENTS	With great interest I have red the manuscript regarding the association between CTEPH and splenectomy. I have same remarks: In the Discussion section the authors did not discuss its clinical relevance. What should we do with this information in our clinical practice? Should we yearly screen all patients after splenectomy on the presence of CTEPH? Please include a paragraph in the Discussion section. Secondly, at the Discussion section the authors stated that CTEPH is a consequence of thrombosis instead of embolism. Do they mean in-situ thrombosis, otherwise on my opinion it is a combination of both: thrombo-embolism. Please clarify. At the Introduction section, the authors described the definition of CTEPH based on a relatively old reference. Please update the reference to: Kim, Delcroix, et al. CTEPH Eur Resp J 2019. The authors stated that the European guidelines on PH showed a 3.4% rate of splenectomy in CTEPH. Please refer to the original article and not the PH guideline in this case. Please uniform the presentation of the authors in all tables. For example: initial(s) and last name throughout all tables. The grammar and English needs editing.

REVIEWER	Arpan A. Sinha, MD

	Jimmy Everest Section of Pediatric Hematology/Oncology, The University of Oklahoma Health Sciences Center, Oklahoma City, OK
REVIEW RETURNED	21-May-2020

GENERAL COMMENTS	Comments to the Author: The authors present a systematic review and meta-analysis demonstrating 4.6% prevalence of splenectomy in patients diagnosed with CTEPH and suggest an association of splenectomy with CTEPH. Comments - General: There are many sentences that are difficult to follow, and the manuscript has many grammatical and spelling errors. The manuscript will benefit tremendously from English editing, grammar, and spelling corrections. Many sentences need to be reworded and revised. Some specific examples from the first paragraph of the abstract: - Whether undergone splenectomy will increase the risk of chronic thromboembolic pulmonary hypertension (CTEPH) remain unclear. Can be revised as “Whether splenectomy increases risk of chronic thromboembolic pulmonary hypertension (CTEPH) remains unclear.” - We hold a systematic review and meta-analysis to explore the association between splenectomy and CTEPH. Can be revised as “We conducted a systematic review and meta-analysis to explore the association between splenectomy and CTEPH” The entire abstract and manuscript will have to checked for correctness and needs to be revised appropriately. Some references need to be corrected. Comments - Introduction: Page 4 – Lines 22-27 – Needs reference. Pathophysiology can also include small vessel disease. Lines 27-33 – Reference for definition does not seem to be correct. Lines 33-38 – Sentence is unclear. From the abstract of the manuscript referenced – 16 out of 49 patients died which is not 90%. Lines 43-45 – Splenectomy is associated with thrombocytosis and not thrombocytopenia. Both references 4 and 5 do not appear to be accurate. Lines 46-53 – Not very clear. It will be helpful to have a few sentences on the pathophysiology of splenectomy and pulmonary hypertension. Lines 56-59 – This sentence is not clear. Splenectomy was one of the factors amongst others thought to contribute to the transformation of PE into a fibrotic vascular occlusion. Comments- Methods: Page 6 - Control group should be specified – unclear what it means in the methods section. Was it the same/similar control groups in each study or were they different? Lines 55-56 – the data appears to be objective data- not clear why it would result in disagreements. Comments – Results and Discussion: 1. How does this study contribute to the understanding of CTEPH and splenectomy from what is already known?

	Prior studies have shown an association between CTEPH and splenectomy and reviewed pathophysiological mechanisms. The meta-analysis does provide a prevalence rate, however as the authors have pointed out many of the studies are small retrospective studies. Also, many hematological disorders responsible for splenectomy are confounding factors for CTEPH. The study can be strengthened by analyzing splenectomy and CTEPH with causes of splenectomy – hematological vs non-hematological conditions vs trauma etc. 2. There is redundancy in these sections (page 9 – lines 35-48 and page 10- lines 7-17- both reporting prevalence rates. The prevalence rate is again repeated on page 12 twice). 3. Page 12- line 20 to page 13-line 4 – The meaning of the paragraph is not clear. If trying to explain the pathophysiology behind splenectomy and thrombosis/CTEPH – it should be better organized with the various theories and processes in different underlying conditions – for e.g. pathophysiology in various hemolytic disorders, role of microparticles, role of thrombocytosis, and coagulation cascade etc.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:
Comments:
1. In the Discussion section the authors did not discuss its clinical relevance. What should we do with this information in our clinical practice? Should we yearly screen all patients after splenectomy on the presence of CTEPH? Please include a paragraph in the Discussion section.
Response: Thank you for your valuable advice. According to the reviewer's suggestion, we have added the clinical relevance at the third paragraph in the discussion section and please see in blue font in the discussion.

2. Secondly, at the Discussion section the authors stated that CTEPH is a consequence of thrombosis instead of embolism. Do they mean in-situ thrombosis, otherwise on my opinion it is a combination of both: thrombo-embolism. Please clarify.
Response: Thank you for your kind comments. They mean both in-situ thrombosis and embolism, and we have revised the expression in the manuscript to make it more intuitive. And it was shown in blue font at the second paragraph in "Discussion" section. (Paragraph 2, line 210)

3. At the Introduction section, the authors described the definition of CTEPH based on a relatively old reference. Please update the reference to: Kim, Delcroix, et al. CTEPH Eur Resp J 2019.
Response: Thank you for your kind suggestion. We have revised the definition of CTEPH and updated the reference at the first paragraph in the Introduction section in the revised revision. (Paragraph 1, line 82-84)

4. The authors stated that the European guidelines on PH showed a 3.4% rate of splenectomy in CTEPH. Please refer to the original article and not the PH guideline in this case.
Response: Thank you for your kind comments. But there was no reference for the 3.4% rate of splenectomy in CTEPH in the European guidelines. And we did not find the original article for the 3.4% rate of splenectomy in CTEPH. We revised the prevalence of splenectomy in CTEPH and added the original article. (Introduction section, Paragraph 2, line 94-95)

5. Please uniform the presentation of the authors in all tables. For example: initial(s) and last name throughout all tables.
Response: Thank you for your careful work. We have re-checked our tables and revised the presentation of the authors and other information.

6. The grammar and English needs editing.
Response: Thank you for reading our manuscript carefully. We are very sorry for our grammar mistakes. We have re-checked our errors throughout. We have sent our manuscript to MogoEdit, a language editing company, to improve the article for language and style. And have provided the certificate in supplementary material confirming that language editing has been performed.

Reviewer 2:
Comments - General:
There are many sentences that are difficult to follow, and the manuscript has many grammatical and spelling errors. The manuscript will benefit tremendously from English editing, grammar, and spelling corrections. Many sentences need to be reworded and revised.
Some specific examples from the first paragraph of the abstract:
- Whether undergone splenectomy will increase the risk of chronic thromboembolic pulmonary hypertension (CTEPH) remain unclear. Can be revised as “Whether splenectomy increases risk of chronic thromboembolic pulmonary hypertension (CTEPH) remains unclear.”
- We hold a systematic review and meta-analysis to explore the association between splenectomy and CTEPH. Can be revised as “We conducted a systematic review and meta-analysis to explore the association between splenectomy and CTEPH”
The entire abstract and manuscript will have to check for correctness and needs to be revised appropriately. Some references need to be corrected.
Response: Thank you for reading our manuscript carefully. We are very sorry for our spelling-errors and grammatical mistakes. We have re-checked our errors throughout. We have sent our manuscript to MogoEdit, a language editing company, to improve the article for language and style. And have provided the certificate in supplementary material confirming that language editing has been performed.

Comments-Introduction:
Page 4–
Lines 22-27–Needs reference. Pathophysiology can also include small vessel disease.
Response: Thank you for your kind suggestion. We have revised definition and updated the reference in the manuscript. (Page 4, lines 81-84)
Lines 27-33–Reference for definition does not seem to be correct.
Response: Thank you for your good comment. We definitely agree with you. We have deleted the sentence in the manuscript. (Page 4, lines 84-87)
Lines 33-38–Sentence is unclear. From the abstract of the manuscript referenced–16 out of 49 patients died which is not 90%.
Response: Thank you for your valuable advice. We have revised this sentence and reference. (Page4, lines 87-88)
Lines 43-45–Splenectomy is associated with thrombocytosis and not thrombocytopenia. Both references 4 and 5 do not appear to be accurate.
Response: We definitely agree with you. We think our references are not very accurate. We have deleted the sentence and references in the manuscript. (Page 4, lines 87-88)
Lines 46-53–Not very clear. It will be helpful to have a few sentences on the pathophysiology of splenectomy and pulmonary hypertension.
Response: Thank you for your kind suggestion. We have revised a few sentences on the pathophysiology of splenectomy or pulmonary hypertension. (Page 4, lines 84-87, Page 5, lines 91-

94)
Lines 56-59–This sentence is not clear. Splenectomy was one of the factors amongst others thought to contribute to the transformation of PE into a fibrotic vascular occlusion.
Response: Thank you for your careful work. We have revised this sentence. (Page 5, lines 91-94)

Comments-Methods:
Page 6-
Control group should be specified–unclear what it means in the methods section. Was it the same/similar control groups in each study or were they different?
Response: Thanks for your advice. We did not restrict the control group. So we deleted this sentence. And we revised and explained the control groups as idiopathic pulmonary arterial hypertension and pulmonary thromboembolism patients in our paper. (Page 6, lines 125)
Lines 55-56–the data appears to be objective data-not clear why it would result in disagreements.
Response: Thanks for your suggestions. Cochrane handbook showed, as a minimum, information that involves subjective interpretation and information that is critical to the interpretation of results (e.g. outcome data) should be extracted independently by at least two people. And at the data extraction stage, we can't avoid data extraction errors and omission. So we need the third author to resolve disagreements of the data extracted between two reviewers.

Comments–Results and Discussion:
1. How does this study contribute to the understanding of CTEPH and splenectomy from what is already known?
Prior studies have shown an association between CTEPH and splenectomy and reviewed pathophysiological mechanisms. The meta-analysis does provide a prevalence rate, however as the authors have pointed out many of the studies are small retrospective studies. Also, many hematological disorders responsible for splenectomy are confounding factors for CTEPH. The study can be strengthened by analyzing splenectomy and CTEPH with causes of splenectomy–hematological vs non-hematological conditions vs trauma etc.
Response: Thanks for your suggestions. There are many confounding factors for CTEPH, but only two trials reported the causes of splenectomy, and we cannot do subgroup analysis to explore the confounding factors for the limited information. This is a limitation for this study, and we added it in the limitation section in the revised revision. (Page 11, lines 238-240)
2. There is redundancy in these sections (page 9–lines 35-48 and page 10-lines 7-17-both reporting prevalence rates. The prevalence rate is again repeated on page 12 twice).
Response: Thanks for your advice. We have deleted the redundancy sections. (Page 11, lines 222-223)
3. Page 12- line 20 to page 13-line 4–The meaning of the paragraph is not clear. If trying to explain the pathophysiology behind splenectomy and thrombosis/CTEPH–it should be better organized with the various theories and processes in different underlying conditions–for e.g. pathophysiology in various hemolytic disorders, role of microparticles, role of thrombocytosis, and coagulation cascade etc.
Response: Thank you for your valuable advice. We have revised the explanation of pathophysiology behind splenectomy and thrombosis/CTEPH. (Page 10, lines 210-214)